

SOMIK GHOSH

Senior AI Engineer | GenAI Engineer | AI Architect | Data & AI Platform Architect

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EXECUTIVE SUMMARY

Senior Data & AI Platform Engineer with 15 years of experience designing enterprise-scale data, analytics, and AI-enabled platforms across banking, healthcare, retail, insurance, and digital media. Currently Engineering Specialist / Staff Engineer at Commonwealth Bank of Australia, leading AI-ready banking data platform initiatives and implementing RAG-based knowledge retrieval patterns for regulated financial services use cases.

Strengths include cloud data engineering, platform modernization, LLM-ready architecture, intelligent workflows, data governance, Responsible AI controls, stakeholder management, and scalable analytics platforms. Led teams of 15+ engineers, improved processing throughput by 70%, increased ETL efficiency by 40%, reduced deployment timelines from weeks to hours, supported platforms processing 10TB+ daily data and 500M+ monthly records, and contributed to \$2M+ annual infrastructure savings.

CORE COMPETENCIES

AI, GENAI & RAG

RAG implementation; LLM integration patterns; OpenAI; Azure OpenAI; AWS Bedrock; prompt engineering; chatbots; AI applications; intelligent workflows; AI automation; enterprise knowledge retrieval; semantic search; hybrid search; citation-aware responses; grounded outputs; vector database integration.

DATA ENGINEERING

Enterprise data platforms; data lakehouse architecture; ETL/ELT modernization; Spark/PySpark; Kafka; Airflow; Snowflake; Redshift; Databricks; Informatica; Teradata; SQL; Python; feature engineering; ML-ready data pipelines.

AGENTIC AI & AI-ASSISTED ENGINEERING

LangChain; LangGraph; LlamaIndex; MCP / Model Context Protocol; tool calling; function calling; agentic workflow design; human-in-the-loop workflows; Cursor; Claude; AI-assisted coding; rapid prototyping; productivity-focused vibe-coding workflows.

CLOUD, PLATFORM & GOVERNANCE

AWS data platforms; cloud migration; Terraform; CI/CD; GitOps; Docker; Kubernetes; data quality; data lineage; metadata management; PII controls; GDPR; HIPAA; SOX; Responsible AI foundations; regulated architecture.

PROFESSIONAL EXPERIENCE

Engineering Specialist / Staff Engineer

Commonwealth Bank of Australia Apr 2025 - Present

- Lead architecture and delivery of AI-ready banking data platforms supporting customer insights, risk analytics, forecasting, regulatory reporting, and intelligent decisioning across financial services operations.
- Implement RAG-based knowledge retrieval patterns for banking data and analytics use cases, covering governed ingestion, context preparation, retrieval workflows, metadata controls, and trusted response generation for internal decision-support scenarios.
- Design data foundations for LLM and GenAI adoption by improving data quality, lineage, access control, reusable data products, automated quality gates, and compliant consumption patterns.
- Partner with banking, analytics, risk, compliance, and engineering stakeholders to convert requirements into scalable data and AI platform roadmaps.
- Use Cursor and Claude for AI-assisted engineering workflows, including rapid prototyping, code generation, SQL/Python optimization, architecture exploration, debugging, documentation, and productivity-focused vibe coding.

Senior Manager - Data Engineering

Tredence Inc. Nov 2024 - Mar 2025

- Led production-grade big data and AI-enablement pipelines on AWS, increasing processing throughput by 70% through Spark optimization, workload tuning, and scalable platform design.
- Migrated legacy ETL systems to Databricks lakehouse architecture, improving pipeline efficiency by 40% and enabling unified batch, streaming, analytics, and ML feature workloads.
- Managed 15 data engineers delivering enterprise BI platforms, ML feature pipelines, operational dashboards, and AI-ready datasets for Fortune 500 clients.
- Implemented GitOps CI/CD, Infrastructure-as-Code, automated testing, and release controls, reducing deployment lead time from weeks to hours with 99.9% release success.
- Optimized cloud resource provisioning and processing architecture, contributing to \$2M+ annual infrastructure savings.

Technical Architect Manager

Impetus Technologies Feb 2023 - Oct 2024

- Architected mission-critical enterprise data platforms processing petabyte-scale workloads, enabling advanced analytics, forecasting, AI/ML integration, and executive decision intelligence.
- Designed AWS Well-Architected data and AI platform layers across ingestion, transformation, serving, analytics, governance, and ML integration using Kinesis/MSK, Glue/EMR, Athena, and QuickSight.
- Established governance standards for metadata, lineage, PII controls, cataloging, GDPR/SOX compliance, and trusted data access required for Responsible AI adoption.
- Engineered federated data integration across 50+ sources including APIs, SaaS platforms, and legacy databases, improving trusted enterprise data availability for analytics and automation.
- Led architecture reviews, stakeholder workshops, roadmap planning, and multi-disciplinary engineering delivery aligned to business KPIs and platform modernization goals.

Technical Architect / Associate Technical Architect

Carelon Global Solutions / Elevance Health Apr 2021 - Jan 2023

- Designed HIPAA-compliant healthcare data fabrics integrating claims, clinical, pharmacy, and socio-economic datasets for population health analytics, risk stratification, provider analytics, and AI/ML-enabled decisioning.
- Led IngenioRx PBM analytics architecture supporting adjudication logic, formulary compliance, rebate modeling, provider network optimization, forecasting, and competitive intelligence workflows.
- Delivered ML-enabled personalization and lifecycle analytics capabilities that improved member retention KPIs by 25%.
- Established reusable healthcare data product patterns using microservices, Kafka event streaming, and domain-driven design, reducing time-to-market for new analytics and AI use cases by 60%.

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PROFESSIONAL EXPERIENCE CONTINUED

Senior Consultant - Data Engineering	Deloitte Consulting	Oct 2015 - Apr 2018
- Led ETL and data architecture for Fortune 100 transformation programs, delivering 100+ Informatica workflows processing 10TB+ daily volumes with 99.99% data accuracy. - Conducted executive workshops and converted ambiguous business goals into technical blueprints, data models, test strategies, and production-grade analytical solutions. - Mentored 12 junior consultants across data engineering, delivery practices, stakeholder communication, and consulting methods, contributing to 90% internal promotion within 18 months.		
Software Associate	Cognizant Technology Solutions	Jun 2014 - Sep 2015
- Developed production Informatica ETL pipelines for telecom billing data warehouses, processing 500M+ monthly records using partitioning, incremental loads, and error recovery patterns.		
Technical Analyst	IBT Media	Aug 2011 - May 2014
- Designed star-schema data marts and analytics pipelines for digital media platforms tracking 1B+ monthly impressions across 50+ KPIs.		

EDUCATION

Bachelor of Computer Applications, Techno India University, West Bengal, India | 2011

CERTIFICATIONS

- AWS Certified Data Analytics - Specialty | Amazon Web Services Training and Certification | Credential visible via Credly.
- Databricks Academy Accreditation - Generative AI Fundamentals | Databricks Badges / credentials.databricks.com.
- HackerRank Silver Badge - Python | HackerRank.

TOOLS & TECHNOLOGIES

AI / GenAI: RAG, LLM Integration, Prompt Engineering, AI Applications, Chatbots, Intelligent Workflows, AI Automation, Responsible AI, AI Governance, OpenAI, Azure OpenAI, AWS Bedrock, Anthropic Claude.

Agentic AI: LangChain, LangGraph, LlamaIndex, MCP, Tool Calling, Function Calling, Agentic Workflows, Human-in-the-Loop, AI Workflow Orchestration.

Vector / Search: Vector Databases, Pinecone, Weaviate, Qdrant, ChromaDB, pgvector, Azure AI Search, OpenSearch Vector Search, FAISS, Semantic Search, Hybrid Search.

AI-Assisted Engineering: Cursor, Claude, AI-assisted coding, vibe coding, rapid prototyping, code generation, architecture exploration, debugging, documentation automation.

Cloud/Data/DevOps: AWS, Databricks, Snowflake, Redshift, EMR, Glue, Kinesis, MSK, Athena, DynamoDB, Spark, PySpark, Kafka, Airflow, Informatica, Teradata, Python, SQL, PL/SQL, Docker, Kubernetes, Jenkins, GitLab CI/CD, Terraform, GitOps.

Governance/Leadership: GDPR, HIPAA, SOX, PII Controls, Metadata Management, Data Lineage, Atlas, Collibra, Architecture Review Boards, Technical Roadmaps, Stakeholder Management, Team Leadership, Agile, SAFe, Mentoring.